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YM2163

Digital Sound Generator (DSG)

■Summary

The DSG (Digital Sound Generator) uses a CPU with built-in quartz crystals to set the initial pitch and timbre.The NMOS-LSI chip effortlessly produces two-beat and rhythmic tones through waveform memory technology.

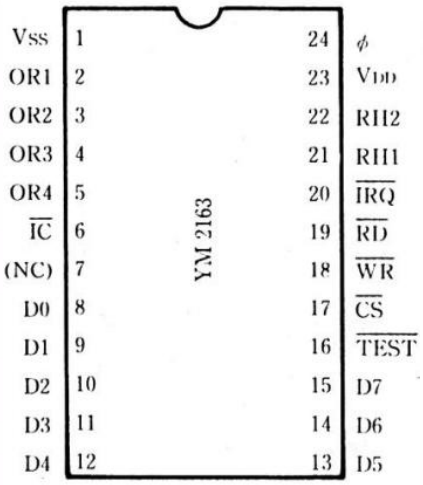
■Features

- 5V Single-Channel Power N-Channel E/D MOS
- TTL 8-bit board
- 8-pin or 16-pin CPU socket for easy connection
- 5 different musical notes and 4 different rhythm sounds, can easily produce 4 octaves scale sounds.
- Built-in 7-bit D/A converter
- Built-in 14-bit timer
- 24-pin DIL socket

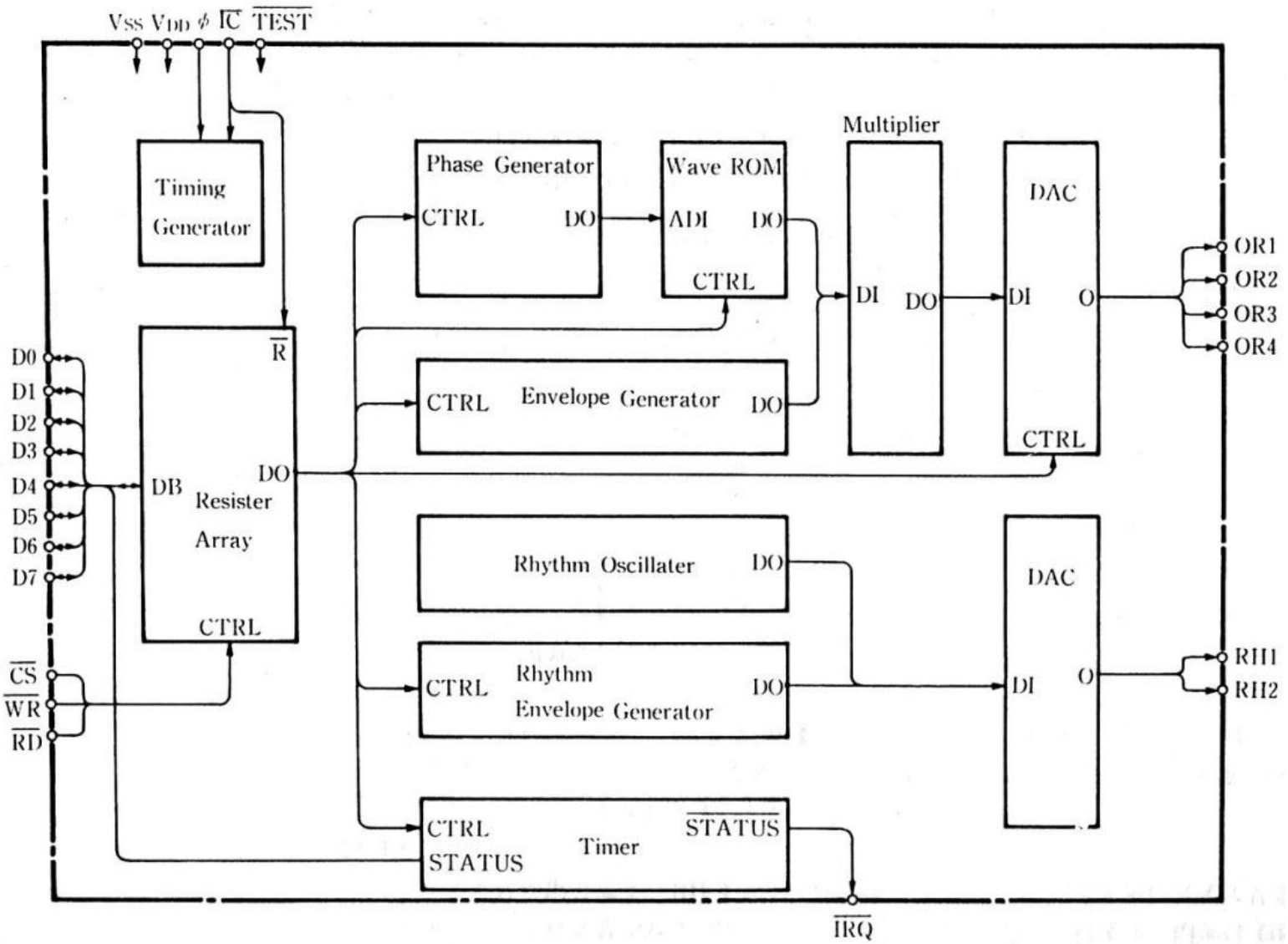
Japan Musical Instrument
Manufacturing Co., LTD

Contents No.LSI-1121630

■Terminal configuration diagram



■Block Diagram



■Terminal Function Description

[1]Vss Ground Terminal

[2]OR1 ~ [5]OR4

The volume output terminal (orchestral sound) features a 4-channel Rex-style structure with analog interface output. This terminal allows independent selection to output signals to either the 4-0 or 4-1 interface.

The internal source circuit design enables mixing with the rhythm sound (output from RH1 and RH2 terminals) through an external load resistor (standard 1KΩ).

[6] $\overline{IC}$

Clear the terminal. When the terminal is in low power mode, it resets the contents of the DSG internal register.

[7] NC

The two terminals inside the DSG are not electrically connected.

[8] D0 ~ [15] D7

Data input terminal. This terminal is used to write data to the DSG internal register array and serves as an output terminal to read the status of the internal timer flag. (See Table 1)

[16] Test

This is a terminal for testing LSI. It contains a full impedance resistor. Do not connect it to anything under normal circumstances.

[17]CS

Select the chip terminal. When this terminal is set to high level, the D0 to D7 data bus terminals enter high interference state, and the data in the DSG internal register array will be overwritten. In this case, the fault flag status cannot be read. (See Table 1)

[18]$\overline{WR}$

Write to the signal input terminal. Switch from CS·WR·RD= "1" to CS·WR= "0"

When a transfer occurs, the path data is written into the register array. (See Table 1)

[19] $\overline{RD}$

Guide signal input terminal. Set CS·WR·RD to '1' to transmit data to the data bus terminal (flag status). This status is output internally by DSG (see Table 1).

	CS	WR	RD	D O ~ D 7 status	Notes
a	1	*	*	high saturation	*: don't care
b	0	1	1	Heinrichs	When a c→b or c→a transfer occurs, the path data is written into the register array.
c	0	0	1	Input (Write)	
d	0	1	0	Output (Read)	Reading the flag status.
e	0	0	0	(prohibit)	Do not use this status.

Table 1 Data Path Control

[20]$\overline{IRQ}$

When the DSG's built-in timer's periodic delay state reverses, the CPU generates an allocation request signal (with '0' activated) and outputs it to the specified terminal.

When the flag register signal (FGR) on the data bus is set to '1', the flag is pulled low, and the IRQ signal returns to '1'. Additionally, the interrupt enable signal (IEN) is set to 0.

The status of the register array is affected by the flag parameter when writing 'O' from the data channel.

The IRQ=1 flag will remain set. (The flag is masked.)

[21] RH1 [22] RH2

The RH1-type drum sound system features two primary tones, each analogically output through the OR1-OR4 terminals. These terminals are equipped with identical S1-S2 interfaces. The load resistor is an external component with a nominal value of 1kΩ.

[23] V_{DD}

Power terminal (nominal voltage +5V)

[24] ϕ

Clock pulse (standard 1 MHz) input terminal.

■ function declaration

All DSG functions are managed by a 23-register array, with the CPU controlling these functions through the register array.

Simply import the data into the register instruction, and the DSG system will automatically generate the sound. According to the configuration of each sound block, the sound will be played in sequence.

●Interval divider: a precision device for adjusting musical intervals.

●Waveform ROM: Stores musical waveforms and supports lossless audio.

● Envelope Generator: Can generate envelope sound effects.

● Multiple tracks: Create sounds with additional envelopes.

● Rhythm oscillator: generates the frequency components of rhythmic sounds.

● Rhythm Envelope Generator: Add low-frequency effects to rhythm sound effects.

●D/A converter: Converts computational results into digital signals, while also digitizing musical and rhythmic sounds.

In addition, the timer generates a CPU 8x frequency division request signal by calculating the period time set in the register array.

As shown in Table 2, the register array stores the following data: When the most significant bit (D7) of the data channel DO ~ D7 is set to 1, the address is written into the DSG's internal register array. If D7 is set to 0, the previously written addresses D0 ~ D6 are cleared. Written data is not overwritten by default (as indicated by the asterisk (*) in Table 2, even when set to' 1 ', it automatically resets to' 0' after a specified period).

;

*Write data

address	D7	D6	D5	D4	D3	D2	D1	D0	COMMENT
80H	0	DV4 (0)	DV3(0)	DV2(0)	DV1(0)	DV0(0)	DV1/2(0)	DV1/4(0)	Weeks (lower)
81H	0	(1)	(1)	(1)	(1)	(1)	(1)	(1)	
82H	0	(2)	(2)	(2)	(2)	(2)	(2)	(2)	
83H	0	(3)	(3)	(3)	(3)	(3)	(3)	(3)	
84H	0	KON(O)	FD(0)	B2 (0)	B1 (0)	DV7(0)	DV6(0)	DV5(0)	Weeks (upper) / octave / Full octaves / Button
85H	0	(1)	(1)	(1)	(1)	(1)	(1)	(1)	
86H	0	(2)	(2)	(2)	(2)	(2)	(2)	(2)	
87H	0	(3)	(3)	(3)	(3)	(3)	(3)	(3)	
88H	0	E2(0)	E1(0)	SUS(0)		W3 (0)	W2 (0)	W1 (0)	Waveform memory / Keep on / Envelope Selection
89H	0	(1)	(1)	(1)		(1)	(1)	(1)	
8AH	0	(2)	(2)	(2)		(2)	(2)	(2)	
8BH	0	(3)	(3)	(3)		(3)	(3)	(3)	
8CH	0	Test 1	VL2(0)	VL1(0)	F4 (0)	F3 (0)	F2 (0)	F1 (0)	Output filter selection / Volume selection / test
8DH	0	2	(1)	(1)	(1)	(1)	(1)	(1)	
8EH	0	3	(2)	(2)	(2)	(2)	(2)	(2)	
8FH	0	4	(3)	(3)	(3)	(3)	(3)	(3)	
90H / 91H / 92H / 93H (1/4)	0	FGR	IEN	HHD ·	HHO ·	SDN	HC ·	BD ·	Rhythm Trigger
									Effective squeeze / Logo Settings / Maintain rank
94H	0		LV4(0)	LV3(0)	LV2(0)	LV1(0)	LV0 (0)	LH (0)	Rhythm Level (HH)
95H	0		(1)	(1)	(1)	(1)	(1)	(1)	(BD)
96H	0		(2)	(2)	(2)	(2)	(2)	(2)	(HC)
97H	0		(3)	(3)	(3)	(3)	(3)	(3)	(SDN)
98H / 99H / 9AH / 9BH (1/4)	0	PT6	PT5	PT4	pT3	PT2	PT1	PT0	Timer period (lower bit)
9 CH / 9 DH / 9 EH / 9 FH (1/4)	0	PT13	PT12	PT11	PT10	PT9	PT8	PT7	Timer period (upper)

Music (Wind and String) / Section Sound / Timer

Note: Add characters (0) to (3) indicate the pronunciation channel.

* read data

address	D7	D6	D5	D4	D3	D2	D1	D0	COMMENT
don’ t care								FLAG	state

Table 2 DSG Address Mask

(1) Frequency and Base Ion • Disable data settings (addresses 80H ~ 87H)

When the number of weeks is set to DV7 ~ DV1/4 and the octave values are B1, B2, the fundamental frequency f

$$f = \frac{\phi/4}{\left(\sum_{i=0}^{7} DVi\right) + DV\,1/2 + DV\,1/4} \times \left(\frac{1}{2}\right)^{(3 - B1 - 2 \times B2)}$$

This will be determined here.

Weeks									
superior			inferior						
DV7	DV6	DV5	DV 4	DV 3	DV 2	DV1	DV0	DV1/2	DV1/4
128	64	32	16	8	4	2	1	1/2	1/4

10 bit ↓ 1024

ϕ: DSG main clock frequency

B2	B	short-cut process
0	0	1
0	1	2
1	0	3
1	1	4

In KON's data configuration, rhythm is controlled by the onset and offset of syllable pronunciation. FD, however, employs a decay mechanism independent of KON.

KON	
0	Key Off (Start attenuation)
1	Press to start (sound starts)

FD	
0	
1	Shape-Shifting Damper (Forced Damping)

(2) Configuration of waveform memory envelope (addresses 88H to 8BH)

Sound waves and envelopes will be selected based on the following data.

wave memorizer			
W3	W2	W1	
0	0	0	
0	0	1	St （string music）
0	1	0	or (organ)
0	1	1	Cl （clarinet）
1	0	0	Pf (Xianuo)
1	0	1	Hc (half fourth)
1	1	0	
1	1	1	

(Refer to Figure 1)

Envelope Selection		
E2	E1	
0	0	Envelope 0 (attenuated sound)
0	1	Envelope 1 (sustained tone)
1	0	Envelope 2 (sustained tone)
1	1	Envelope 3 (sustained tone)

(Refer to Figure 2)

Continue to specify	
SUS	
0	Keep Off
1	Keep on

(Refer to Figure 2)

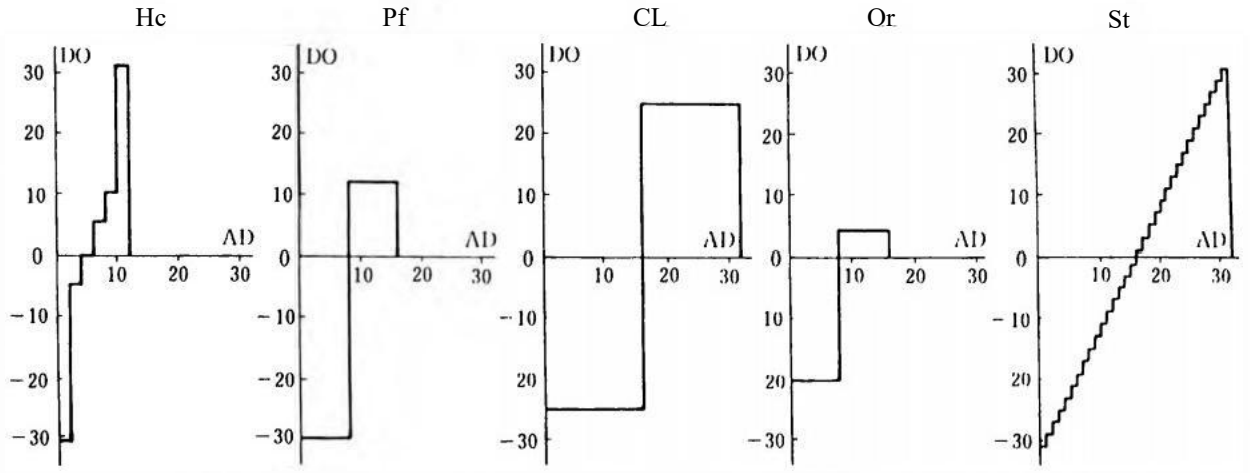


Figure 1 Waveform Memory (Melody)

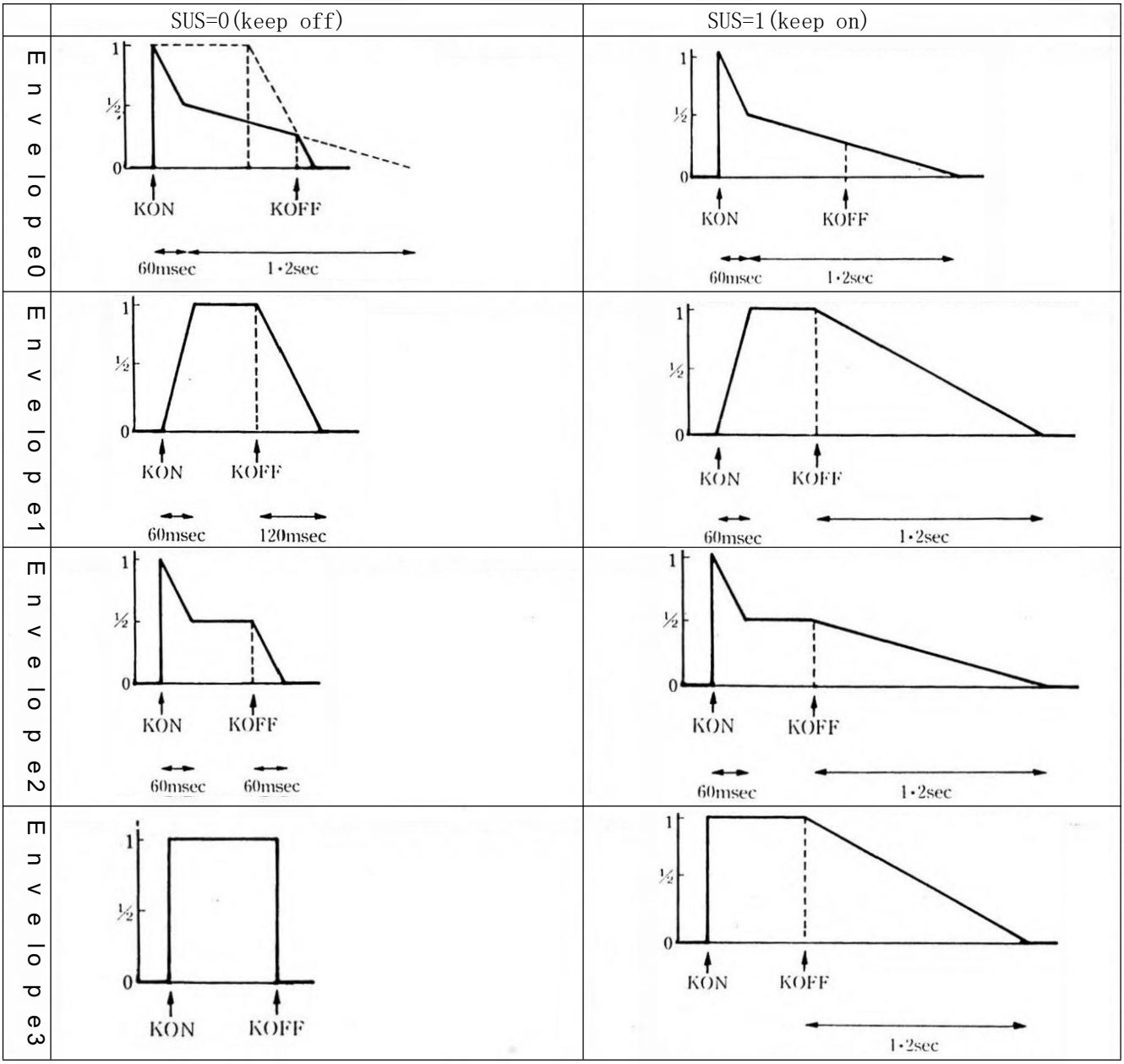


Figure2 Envelope (Melody)

(3) Volume and Filter Selection Settings (Addresses 8 CH to 8 FH)

Volume adjustment data and output terminal designation data will be set as follows.

Volume (water chime)		
V2	V1	
0	0	0 dB (normal)
0	1	-6dB
1	0	-12dB
1	1	-∞dB

Output terminal selection (filter)		
F1	0	Do not output to OR1 terminal.
	1	Output to OR1 terminal.
F2	0	Do not output to OR2 terminal.
	1	Send the output to the OR2
F3	0	Do not output to the OR3 terminal.
	1	Output to OR 3 terminal.
F4	0	Do not output to OR4 terminal.
	1	Output to OR4 terminal.

(4) Rhythm Trigger (Addresses 90H to 93H)

When the data "1" is deviated, the system will emit the following rhythm sound effect, and the built-in rhythm envelope generator will naturally change, and the data will automatically recover to "0".

Rhythm Tracker			explain
BD	0		Output the signal to terminal RH1.
	1	Ding	
HC	0		Output the signal to the RH1 terminal.
	1	Speak at high volume	
S D N	0		Output the signal to the RH2 terminal.
	1	Military drum noise	
HHO	0		Output the signal to the RH2 terminal.
	1	High Hat (Open) Pronunciation	
HHD	0		Output the signal to the RH2 terminal.
	1	Hat (end) pronunciation	

(5) Rhythm hierarchy (address 94H ~ 97H)

When controlling the rhythm output rate through the DSG external device, the rate conflicts with the built-in rhythm envelope generator. In this case, the

Rhythm Level				
LV4	LV3	LV2	LV1	LV0
16	8	4	2	1

Level hold	
LH	
0	Do not keep the set rhythm level
1	Maintain the set rhythm level

e is typically set to 1.

The volume level ranges from 0 (maximum) to 31 (minimum).

(6) Timer interrupt control (addresses 98H-9F H, 90H-93H)

The DSG's internal interrupt flag is set every time the preset timing period is reached.

$$\text{时间周期PT} = (1 + \sum_{i=0}^{13} \text{PTi}) \times \frac{28}{\phi} \quad \text{(second)}$$

Here, ϕ: DSG's main clock frequency (Hz)

superior:							F Position						
pT13	P T12	PT11	PT10	PT9	PT8	PT7	PT6	PT5	PT4	PT3	PT2	PT1	PT0
8192	4096	2048	1024	512	256	128	64	32	16	8	4	2	1

The interrupt flag is reset when the reset signal FGR= "1" is written, and FGR automatically returns to "0" at this time.

When the distribution completion flag is in an interrupted state, the IEN signal will turn on. In this case, the $\overline{\text{IRQ}}$ terminal will output an inverted signal, and the D0 terminal will also output a corresponding signal in read mode.

■ **electricalcharacter**

Absolute maximum freeze

mark		min	sta	max	unit
VI	terminal voltage	-0.3		7	V
Top	operating temperature	0		70	℃
Tstg	Save temperature	-50		125	℃

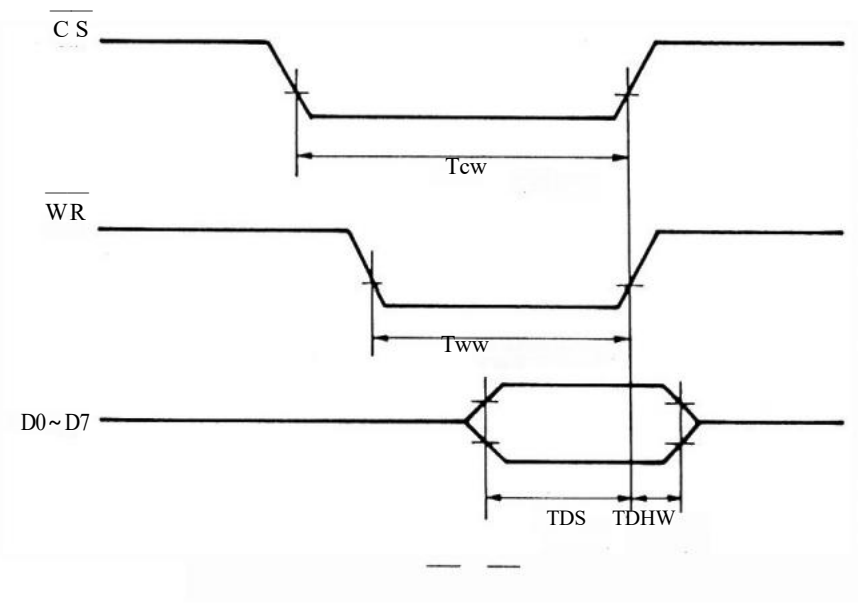
DC Characteristics (@VDD=4.5~5.5V)

mark		min	sta	max	unit
V_{DD}	supply voltage	4.5	5.0	5.5	V
I_{DD}	Power supply current @RL=1KΩ		25	45	mA
V_{IL}	Enter low voltage （ϕ、$\overline{CS}$、D0-D7、$\overline{WR}$、$\overline{RD}$、$\overline{IC}$）			0.8	V
V_{IH}	Enter high-level voltage （ϕ、$\overline{CS}$、D0-D7、$\overline{WR}$、$\overline{RD}$、$\overline{IC}$）	2.0			V
V_{OL}	Output low voltage (D0, IRQ) @IOL=1.6mA			0.4	V
V_{OH}	Output high-level voltage (D0)	2.4			V
I_{LK}	Enter leakage current （ϕ、$\overline{CS}$、$\overline{D0-D7}$、$\overline{WR}$、$\overline{RD}$、$\overline{IC}$） @VI=5V			10	μA

AC characteristic

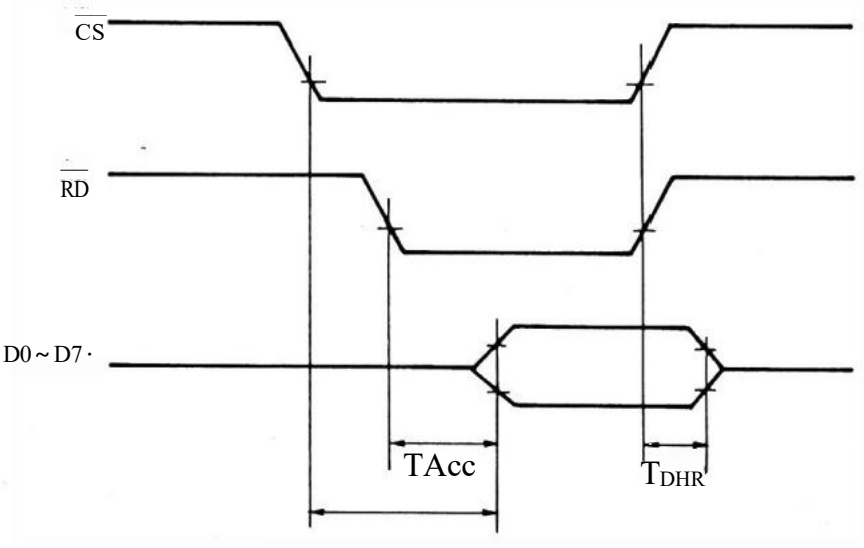
mark			min	sta	max	unit
Tcw	Chip selection in pulse mode	(Figure 3)	135			ns
Tww	In the optical signal pulse	(Figure 3)	135			ns
T_{DS}	Data Set Time	(Figure 3)	160			ns
T_{DHW}	Data retention time (write)	(Figure 3)	30			ns
T_{ACC}	Access Time	(Figure 4)			400	ns
T_{DHR}	Data retention time (read)	(Figure 4)	0			ns
f_{ϕ}	Clock frequency	(Figure 5)	0.9	1.0	1.1	MHz
$T_{\phi ON}$	Clock enable time (@fϕ=1MHz)	(Figure 5)	400			ns
$T_{\phi OFF}$	Clock shutdown time (@fϕ=1MHz)	(Figure 5)	400			ns
$T_{\phi R}$	Clock up time	(Figure 5)			80	ns
$T_{\phi F}$	Clock down time	(Figure 5)			80	ns
T_{OD}	Output delay time $\overline{IRQ}$	(Figure 5)			700	ns
Vp-p	Maximum Output Vibration (OR1-4, RH1-2) @RL=1KΩ, VDD=5V	(Figure 6)	0.60	0.90	1.40	V

Figure 3: Write time



Note: The TDS and TDHW criteria are determined when either CS or WR reaches a high level.

Figure 4: Read time



Note: TACC CS uses DR the party with lower rank and slower response as the reference.

TDHW is CS based DR on the time when one of the parties reaches the higher level.

Figure 5 Clock/Output Timing

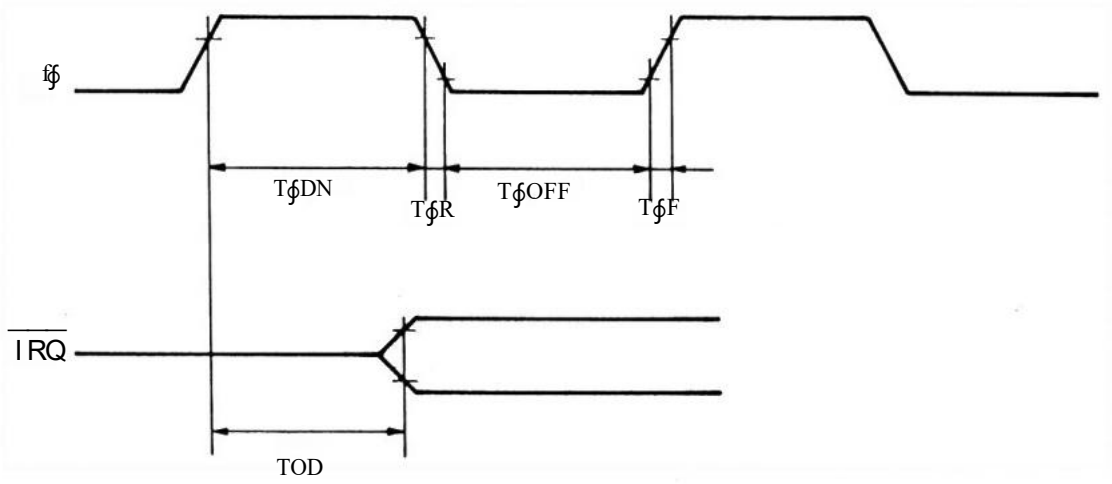
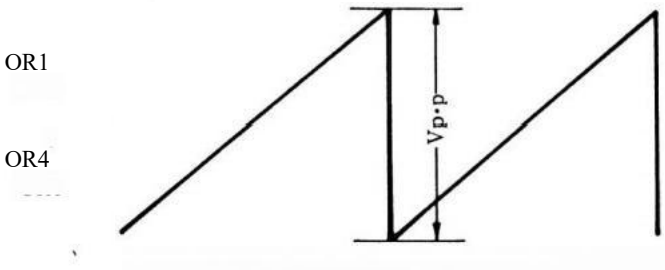
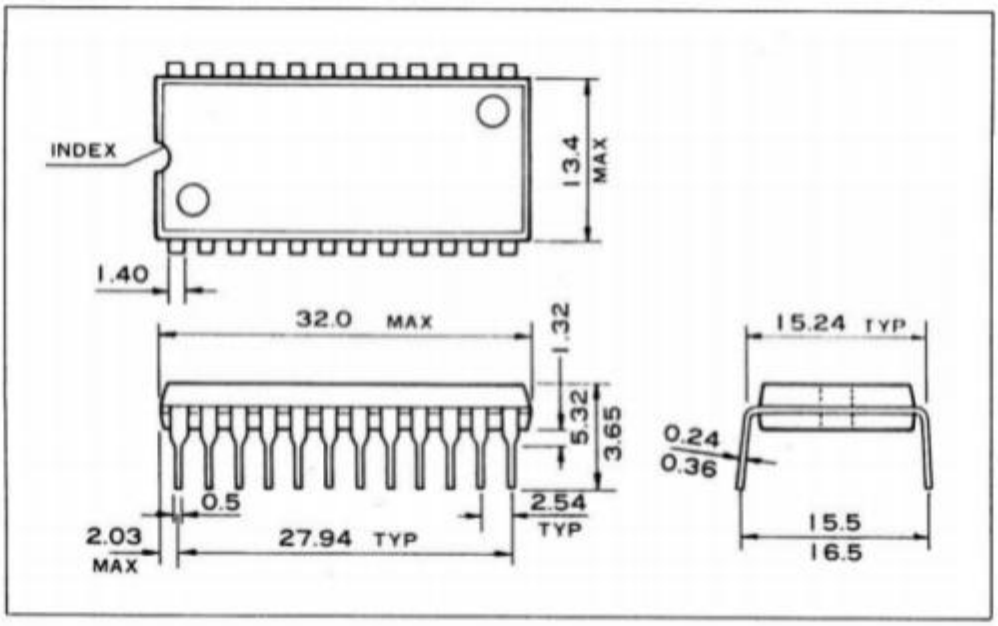


Figure 6 Maximum output amplitude



(Note) When W1=1, W2=W3=0, VL1=0, and VL2=0, this rule applies.

■Dimension



Note: Product specifications are subject to change without notice due to improvements or other reasons。

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